Warranty Claim Fraud Prediction

Project report: exploring 358 AC and TV warranty claims and trying to flag the fraudulent ones

Notebook: warranty-claim-fraud-prediction-final.ipynb | Author: Yash Jain

1. Introduction

Warranty fraud costs manufacturers money, and checking every claim by hand is slow. This project looks at a small dataset of warranty claims for air conditioners and TVs in India. The first goal is to understand what fraudulent claims have in common: where they come from, how they are bought, how large they are, and how the customer behaves on the phone. The second goal is to see whether a classifier can flag fraudulent claims automatically.

The exploratory part gave some clear signals. The modelling part, honestly, did not work: three standard classifiers reached 91 to 92% accuracy, but that is no better than guessing "genuine" for every claim. The report explains why this happened and what I would change.

2. Data

The dataset is a Kaggle warranty claims file (saved as df_Clean.csv) with 358 rows and 20 columns, plus an unnamed index column that I drop. There are no missing values and no duplicate rows. The target is *Fraud* (1 if the claim was fraudulent). 35 of the 358 claims (9.8%) are fraud, so the classes are heavily imbalanced.

Column	Description
Region, State, Area, City	Location of the claim: 8 regions, 20 states, 27 cities, Urban or Rural
Consumer_profile	Business (134 claims) or Personal (224)
Product_category, Product_type	Household / AC (167 claims) or Entertainment / TV (191)
AC_1001, AC_1002, AC_1003, TV_2001, TV_2002, TV_2003	Issue with each of six parts: 0 none, 1 repair, 2 replacement
Claim_Value	Claim value in INR: 0 to 50,000, median 7,194, mean 11,995
Service_Centre	Service centre code, 10 to 16
Product_Age	Product age in days: 3 to 991, median 60
Purchased_from	Dealer (139), Manufacturer (135) or Internet (84)
Call_details	Length of the customer care call in minutes: 0.5 to 30, median 6.5
Purpose	Claim (171), Complaint (146) or Other (41)
Fraud	1 for fraud (35 claims), 0 for genuine (323)

Two things about the structure are worth noting. First, Product_category and Product_type are the same information: every AC is "Household" and every TV is "Entertainment". Second, the three AC

part columns are always 0 for TVs, and the three TV part columns are always 0 for ACs, so a claim only ever has issues on one product line. Ten claims have a claim value of exactly 0.

3. Preprocessing

Cleaning. I dropped the index column and renamed the part-issue codes (0, 1, 2) to the labels "No Issue", "repair" and "replacement" for the plots.

Outlier removal. I removed outliers in Claim_Value with the 1.5 x IQR rule. The quartiles are 4,006 and 15,000, so the upper limit is about 31,491. This removed 24 claims, leaving 334. As discussed in section 6, 8 of the 24 removed claims were fraud.

Encoding. All remaining text columns (region, state, area, city, consumer profile, product category and type, the six issue columns, purchase channel and purpose) were label-encoded to integers.

Split. 70% training and 30% test with random_state 42 and no stratification: 233 training claims (19 fraud) and 101 test claims (8 fraud).

4. Exploratory analysis

A note on how I read the plots. Most of my notebook charts are histograms and count plots of claims split by fraud. They show where the claims are, not where fraud is most likely, and with only 35 fraud cases the bars are tiny. For this report I recomputed the fraud *rate* inside each group from the CSV (Figure 1). Counts are shown in brackets, and the small groups should be treated with care.

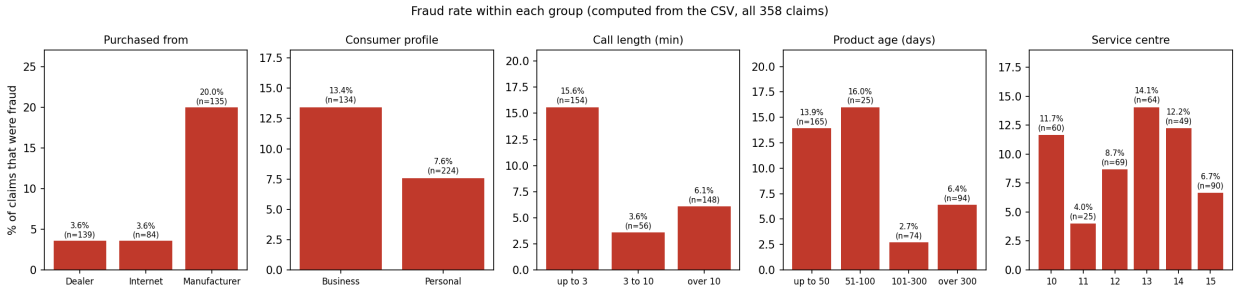


Figure 1. Fraud rate by group, computed from the CSV (not in the original notebook).

Purchase channel

This is the strongest pattern in the data. 27 of the 35 fraud claims (77%) were for products bought directly from the manufacturer, a fraud rate of 20.0%, against 3.6% for dealer and internet purchases. The notebook noted the same thing and called it a concern for the company.

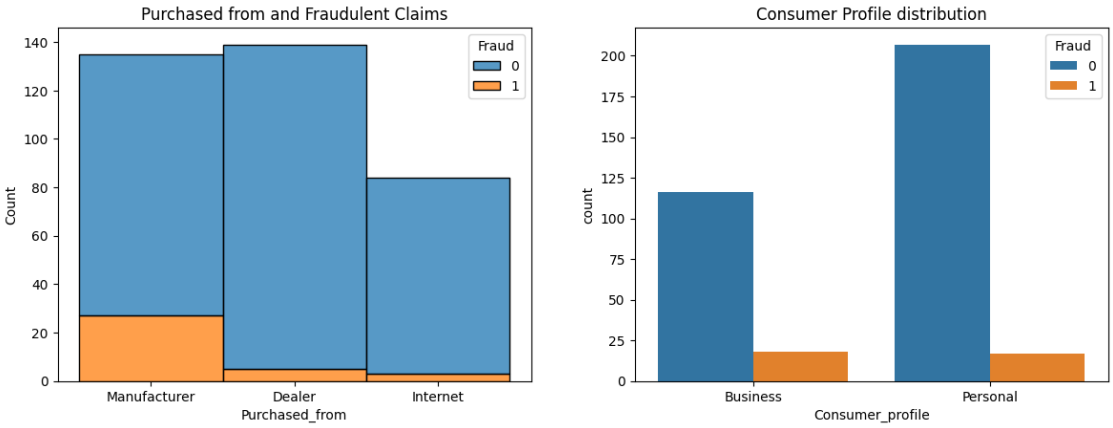


Figure 2. Claims by purchase channel (left) and consumer profile (right), split by fraud.

Customer behaviour

Fraudulent claims come with short calls. 24 of the 35 frauds were on calls of up to 3 minutes (15.6% of those claims), against 5.4% for longer calls. The median call lasts 2.4 minutes for fraud and 7.0 minutes for genuine claims. Fraudulent claims are also on young products: the median product age is 18 days for fraud and 81 days for genuine claims, and the fraud rate is 14 to 16% for products up to 100 days old, against under 7% for older ones. Business customers have a higher fraud rate (13.4%) than personal customers (7.6%).

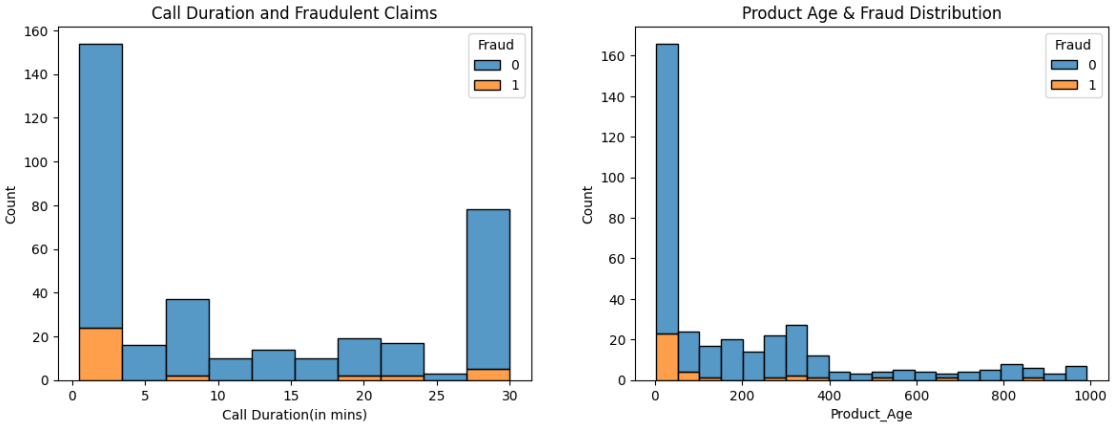


Figure 3. Call duration (left) and product age (right), split by fraud.

Claim value

Fraudulent claims are larger. The median claim is 15,000 for fraud and 7,000 for genuine claims, and the mean is 20,644 against 11,057. The maximum is 50,000 for both groups.

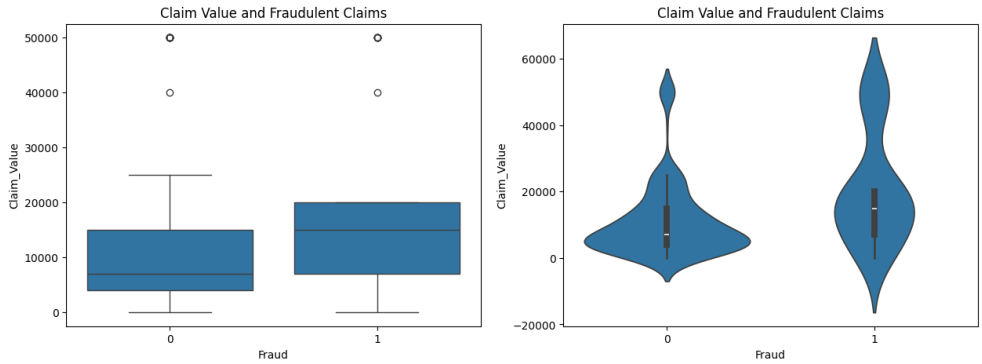


Figure 4. Claim value for genuine (0) and fraudulent (1) claims.

Product, parts and service centre

TV claims have a slightly higher fraud rate (11.0%) than AC claims (8.4%). For the part-issue plots, the notebook observed that AC fraud tends to happen when there is no issue with the part, while TV fraud happens both with and without an issue. The service centres with the highest rates are 13 (14.1%, 9 of 64), 14 (12.2%) and 10 (11.7%), and the lowest is centre 15 (6.7%) among the larger ones. These differences rest on very few cases.

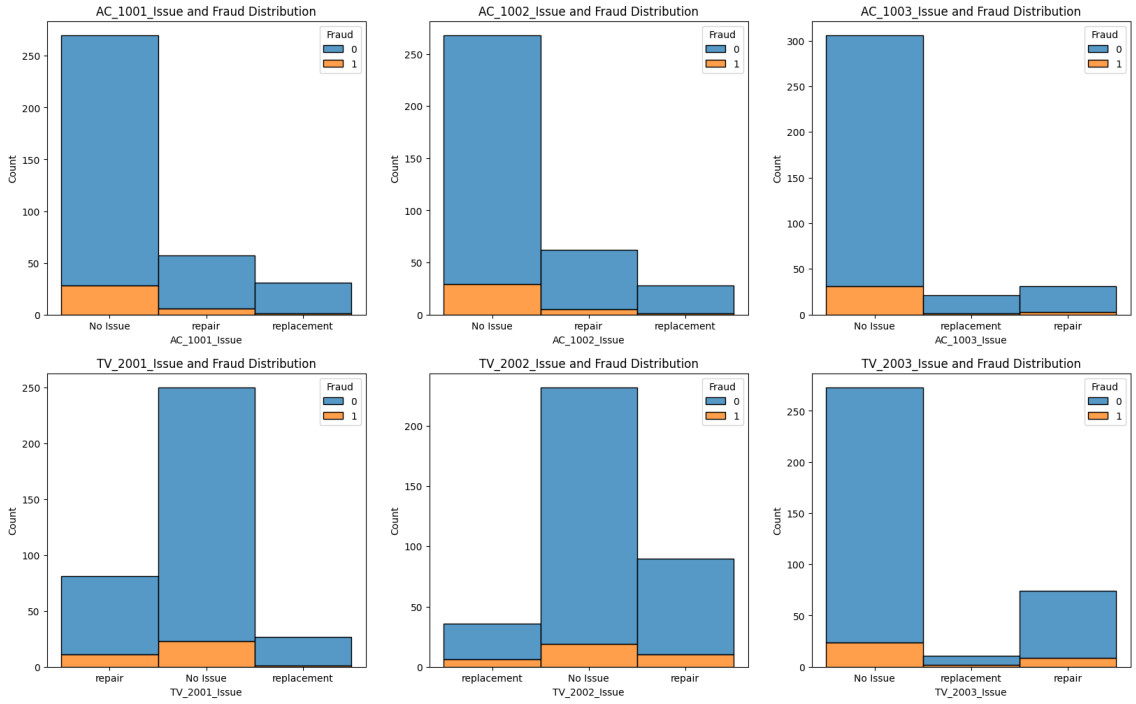


Figure 5. Part issues and fraud for the three AC parts (top) and three TV parts (bottom).



Figure 6. Service centre (left) and purpose of contact (right), split by fraud.

Location

The notebook concluded that fraud is concentrated in the South and in cities like Hyderabad and Chennai, and that urban areas have more fraud than rural ones. The rates tell a different story. Urban and rural areas are identical (9.6% and 10.1%): urban areas simply have more claims. East (17.9%, 7 of 39) and West (14.3%) have the highest regional rates, South is 8.2% and North West has none (0 of 27). No state has more than 7 fraud cases and no city has more than 5, so I would not draw conclusions about individual places.

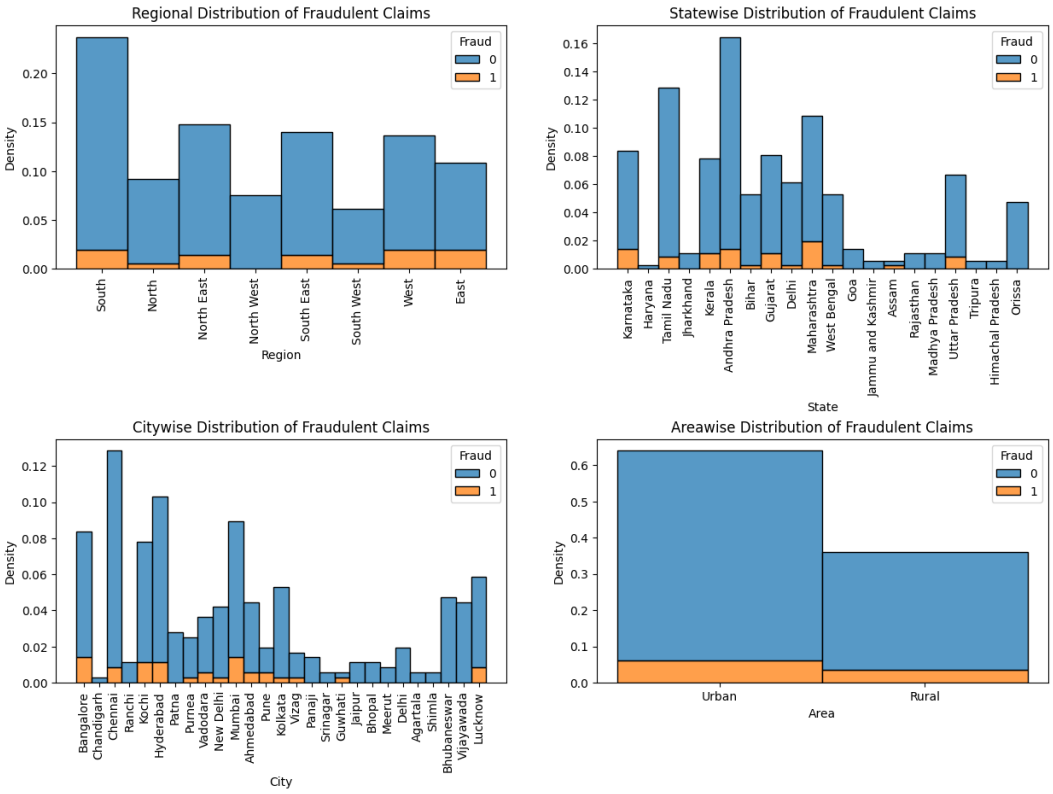


Figure 7. Distribution of claims and fraud by region, state, city and area (as drawn in the notebook).

Correlations

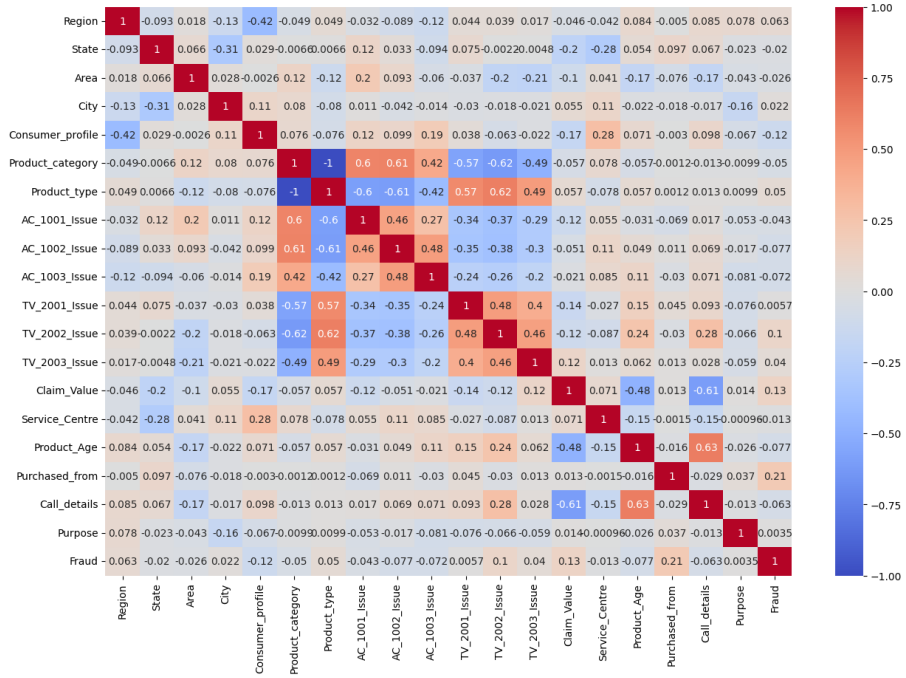


Figure 8. Correlation matrix of the label-encoded data.

5. Modelling

Three classifiers were trained on the training set. For the decision tree and the random forest I used GridSearchCV with 5-fold cross-validation and accuracy as the score. The tree grid had 500 combinations (2,500 fits) over depth, minimum samples per leaf and per split, criterion and random state, and the forest grid had 256 combinations (1,280 fits). Both searches picked a maximum depth of 2, a minimum of 2 samples per leaf and per split, the gini criterion and random_state 0. Logistic regression used its default settings.

I then built the final models. The random forest used the grid-search values. The decision tree, however, was coded with max_depth=4 and not the 2 found by the search. I set that by hand.

6. Results

	Decision tree	Random forest	Logistic regression
Train accuracy	0.931	0.918	0.918
Test accuracy	0.911	0.921	0.921
Precision (fraud)	0.33	0.00	0.00
Recall (fraud)	0.12	0.00	0.00
F1 (fraud)	0.18	0.00	0.00
Frauds found (of 8)	1	0	0
False alarms	2	0	0

The random forest and logistic regression predicted "genuine" for all 101 test claims. A model that always says "genuine" is right 93 times out of 101, which is 92.1%, exactly their accuracy. The decision tree found 1 of the 8 fraud claims and raised 2 false alarms, and its accuracy (91.1%) is below the baseline.

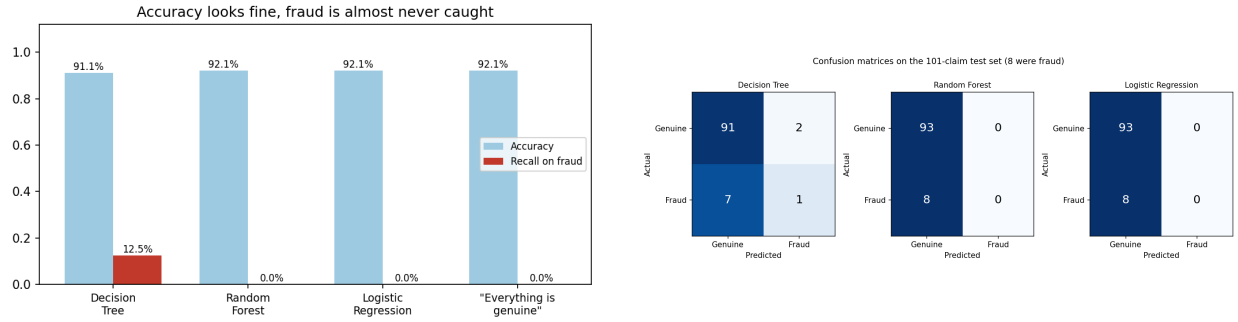


Figure 9. Accuracy against recall for fraud (left) and confusion matrices on the test set (right).

Feature importance. In the decision tree, Claim_Value is the most important feature (0.43), then TV_2002_Issue (0.20), Service_Centre (0.12) and City (0.09). In the random forest, the top features are Claim_Value (0.20), Product_Age (0.18), Call_details (0.12), Purchased_from (0.10) and TV_2002_Issue (0.08). The forest's ranking matches the patterns found in the exploratory analysis, even though the forest itself never predicts fraud.

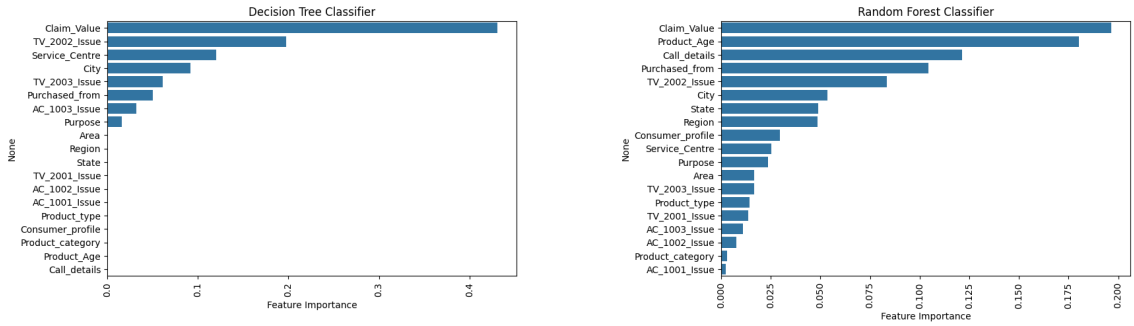


Figure 10. Feature importance for the decision tree (left) and the random forest (right).

7. Limitations

- **The models do not detect fraud.** Two of them never predict it and the third finds one case out of eight. The 91 to 92% accuracy is the same as the "always genuine" baseline. The notebook's conclusion calls this "excellent accuracy".
- **The outlier filter removed fraud.** The IQR rule dropped 24 claims: 8 of the 35 fraud claims (23%) but only 16 of the 323 genuine claims (5%). Since high claim values are one of the clearest signals of fraud, the filter threw away some of the cases the model needed to learn from (Figure 11).
- **Tuned on accuracy.** Scoring the grid search by accuracy favours a model that ignores the minority class, so both searches picked a depth-2 model. Recall, F1 or PR-AUC with stratified folds would be more appropriate. Random_state was also treated as a hyperparameter, which it is not.
- **The final tree does not match the search.** The grid search chose a depth of 2 and I coded a depth of 4. When I re-ran the pipeline with depth 2, the tree also predicted every claim as genuine, so the depth-4 setting is what lets it find a single fraud.
- **Very small test set.** There are only 8 fraud claims in the test set, so each one is worth 12.5 points of recall. There was no stratified split and no cross-validated comparison.
- **Weak location analysis.** My comments about states, cities and urban areas were based on counts. With at most 7 fraud cases per state and 5 per city, those rankings are mostly noise.
- **Encoding and redundant columns.** State, city and region were label-encoded, so the models read them as ordered numbers, and with 358 claims they probably should be dropped. Product_category duplicates Product_type.
- **Metrics that do not fit.** The notebook prints R^2 (negative for all three models) and MSE (which is just the error rate for a 0/1 target).
- **Unknown origin.** I do not know how the data was collected or how claims were labelled as fraud, so I would not assume these patterns hold for real warranty claims.

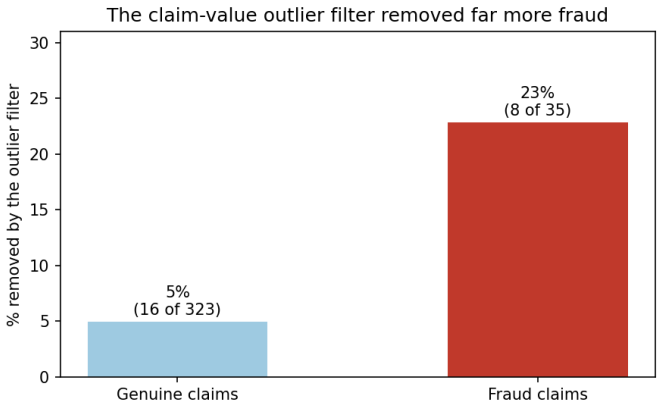


Figure 11. Share of claims removed by the claim-value outlier filter, by class.

8. Conclusion

The exploratory analysis found clear patterns: fraud is far more common for products bought from the manufacturer (20% against 3.6%), on short customer care calls, on young products, with larger claim values, and with business customers. Location matters much less than my notebook suggested.

The classifiers could not use these patterns. With 35 fraud claims out of 358, accuracy-based tuning and a random split produced models that either ignore fraud or find a single case. The honest summary is that this project gives a useful description of fraud in the data, but not a working detector.

9. Future work

- Keep the high-value claims, and judge models by recall, precision, F1 and PR-AUC instead of accuracy.
- Use stratified k-fold cross-validation on all 358 claims, tuned on F1 or recall.
- Handle the imbalance with class weights, oversampling (SMOTE) or a lower decision threshold, and report the trade-off between missed fraud and false alarms.
- Build a small set of simple features: manufacturer purchase, call under 3 minutes, product age under 100 days, claim value above the median.
- Drop the duplicate and structural columns, and one-hot encode what remains (or drop state and city).
- Compare against a simple rule built from those features, which may do as well as the models.
- Collect more fraud examples, and find out how the labels were produced.

10. Tools used

Python, pandas, NumPy, Matplotlib, Seaborn, scikit-learn, Google Colab.